Beta reduction and physical clearance in a parametric booster-to-storage-ring transfer line

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Abstract

A booster-to-storage-ring transfer line must transport the extracted booster beam while satisfying the optical requirements of storage-ring injection. For fixed beam parameters and prescribed entrance and exit optics, quadrupole rematching preserves the nominal beam sizes at both ends. We show for a 21.8 m, 4 GeV line of the Korea Light Source design that the limiting restriction is the injection-septum blade, 1.48 mm from the injected beam in the geometry of a companion nonlinear-kicker study, and that its nominal clearance is independent of the quadrupole settings. What the optics do control is where booster shot jitter arrives: with matched ends the exit jitter depends only on the horizontal phase advance, which exchanges position jitter at the blade for angle jitter at the injection kicker. After trajectory correction, which reduces the loss from 10 % to 1.7 %, this jitter sets the remaining blade loss. A Gaussian loss model agrees with tracking within 1 % and serves as a design objective. Matched solutions reach only a 200°–232° phase-advance interval; a setting at its edge cuts the tracked loss on fresh machines from 2.30 % to 0.69 % at 13 % more exit-angle jitter, whereas lowering the vertical peak beta by 45 % gives no resolved change, and the mean minimum clearance metric cannot resolve either result. A dispersion-free exit is feasible, widens the reachable interval and lowers the predicted loss a further 65-fold. Aperture dimensions other than the blade, and the error levels, are declared assumptions.

Keywords: transfer line; injection septum; beam loss; betatron phase advance; booster jitter; trajectory correction

1 Introduction

Diffraction-limited storage rings require control of the beam distribution delivered to the injection region [1]. The booster-to-storage-ring (BTS) transfer line connects the extracted booster beam to the prescribed injection conditions of the storage ring. Its optical design must accommodate the incoming emittance and momentum spread, satisfy the exit lattice functions, and maintain adequate clearance through the magnets and injection septum. At the septum, the beam envelope and trajectory must fit within a channel bounded by the blade; downstream, both position and angle affect the conditions delivered to the injection kicker. The 4 GeV Korea Light Source (KLS) design [2, 3] provides the application considered here, with the injection geometry specified by a companion nonlinear-kicker magnet (NKM) study [4].

Previous transfer line studies address optical matching, trajectory stability, and injection-region constraints through complementary approaches. The SLS 2.0 study separates quadrupole-scan beam characterization from position and angle control and identifies septum restrictions [5]. The ESRF–EBS transfer-line rematching includes injection-region fields and mechanical constraints [6], while the Sirius injection study distinguishes the injector stages and the effects of incoming beam

size [7]. Pulsed-sextupole injection at MAX IV also connects the delivered beam distribution to storage-ring capture [8]. These studies motivate treating the nominal optical match, clearance at a physical restriction, and sensitivity to incoming trajectory fluctuations as separate design criteria. In particular, a reduction in a beta function is useful for transmission only to the extent that it improves the beam distribution at a restriction that contributes to loss.

A common transfer-line design objective is to reduce the peak beta functions and thereby increase aperture margin. For fixed emittances, momentum spread, and matched entrance and exit lattice functions, however, quadrupole rematching preserves the nominal beam sizes at the boundaries. Restrictions in the fixed sections upstream and downstream of the quadrupoles can therefore limit the line even when its interior envelopes are reduced. The same boundary conditions leave freedom in the betatron phase advance, which changes the transport of booster position and angle jitter. A matched setting can consequently change the loss probability at a septum blade without appreciably changing its nominal clearance. Distinguishing these mechanisms is necessary to determine whether a proposed optics improvement addresses the limiting restriction or only redistributes margin elsewhere in the line.

The goal of this study is to determine how quadrupole rematching affects physical clearance and beam loss in a BTS line with prescribed endpoint optics, and to identify a design criterion sensitive to the remaining loss mechanism. The KLS line is modeled with element-wise apertures and the injection-septum blade geometry of the companion NKM study. Reference and reduced-beta settings are compared after static trajectory correction using field trims of existing dipoles. A Gaussian loss probability derived from the transported beam and centroid covariance is validated against particle tracking and used to select a loss-based setting before evaluation on independently seeded machines. The analysis also examines the reachable phase advance, a dispersion-free exit target, and sensitivity to the assumed error levels. The study ends at the transfer-line exit; storage-ring capture and NKM transparency require the subsequent injection calculation.

The paper is organized as follows. Section 2 defines the BTS lattice, beam covariance, apertures, and trajectory-correction model. Section 3 presents the clearance and loss criteria, matched-end jitter transport, and numerical validation procedures. Section 4 compares the aperture restrictions, corrected trajectories, loss-based setting, exit targets, and error sensitivities. Section 5 discusses the design implications and limitations. The supplementary material provides the four-station aperture comparisons and historical optical statistics.

2 Parametric model and physical observables

2.1 Lattice, coordinates, and boundary conditions

The Accelerator Toolbox (AT)/pyAT model [9, 10] contains 36 elements over 21.789 m, from marker m1 to marker m4 downstream of sept_ex. It has nine 0.2 m quadrupoles in three triplets, three main dipole bending magnets (`b1`, `b2`, and `b3`, in beam order), an extraction kicker and two model septa (Fig. 1). Local coordinates are $(x, p_x, y, p_y, \delta, ct)$ with positions/path length in m, normalized canonical momenta approximately angles in rad, and $\delta = \Delta p/p_0$ dimensionless. The six design bend angles are 7.5 mrad, 88.5 mrad, −111.701 mrad, 176 mrad, −111.701 mrad and 88.5 mrad; reference geometry is held fixed when fields or particle momenta change.

Entrance and exit lattice functions (Table 1) are inherited configured values. The exit m4 is the injection-septum exit: the companion NKM study uses exactly these values as its injected beam and transports it 2.26 m to the kicker. The configured nonzero exit dispersion (80.9 mm, 47.5 mrad) differs from the ring dispersion at the injection kicker, which is about 0.3 mm; we therefore also examine a dispersion-free exit. Quadrupole family bounds, bore radius, pole-field limit, the 60 m beta screening ceiling and square aperture limits are modeling assumptions. The BTS line is independently validated using analytical maps of the same AT element sequence.

The four-station comparison model places square apertures at markers m1–m4 (19.35 mm at the line ends, 30 mm at two interior markers) and none on the magnets. That four-station model

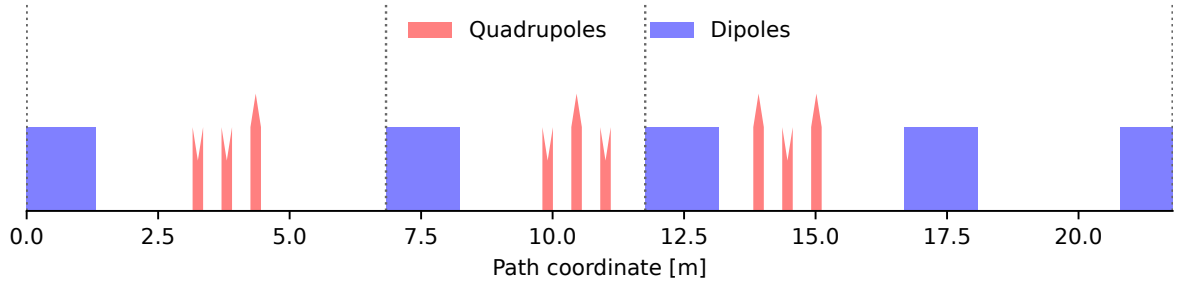


Figure 1: Configured BTS element sequence drawn with Accelerator Toolbox synoptic plotting. Red symbols are quadrupoles and blue blocks are dipoles; blank intervals are drifts. Dotted lines mark the markers m1–m4 used for boundary and intermediate optics observations.

Table 1: Configured entrance and exit targets. The exit values are the injected beam at the injection-septum exit used by the companion NKM study [4]. A dispersion-free exit is examined in Section 4.4.

Observable	m1 entrance	m4 target
β_x, β_y [m]	7.560000, 12.269000	2.336495, 4.256241
α_x, α_y [1]	1.523100, −1.654700	−0.016335, 0.017772
D_x [m], D'_x [1]	0.276200, −0.065700	0.080868, 0.047472

and its hypothetical internal restrictions are documented in the Supplement; this article uses the element-wise restrictions of Section 2.3.

2.2 Beam covariance and clearance

The new study uses the booster horizontal design input $\epsilon_x = 7.67\,\mathrm{nm\,rad}$ and $\sigma_\delta = 0.106\,\%$ [3]. The vertical input $\epsilon_y = 0.767\,\mathrm{nm\,rad}$ is an assumed 10 % ratio. Betatron coordinates and momentum are independent untruncated Gaussians before dispersion is added; initial transverse coupling is zero and $ct = 0$ for this RF-free single pass. The transverse blocks of the complete covariance are

$$\Sigma_{u,\mathrm{bet}} = \epsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & (1+\alpha_u^2)/\beta_u \end{pmatrix}, \qquad \Sigma_0 = \Sigma_{\mathrm{bet}} + \sigma_\delta^2 \mathbf{d}\mathbf{d}^{\mathsf{T}}, \tag{1}$$

where $\mathbf{d} = (D_x, D'_x, D_y, D'_y, 1, 0)^{\mathsf{T}}$. Rolls can generate coupling downstream. Full covariance transport, $\Sigma_s = M_s \Sigma_0 M_s^{\mathsf{T}}$, retains that coupling and momentum correlation. A local affine map gives the centroid $\bar{\mathbf{z}}_s = M_s \bar{\mathbf{z}}_0 + \mathbf{b}_s$.

For each aperture plane with lower/upper edges a_-, a_+, we define

$$c_u(s) = \min[a_+ - \bar{u}(s), \bar{u}(s) - a_-] - 5\sqrt{\Sigma_{uu}(s)}, \qquad c_{\min} = \min_{s \in \mathrm{stations}, u \in \{x,y\}} c_u(s). \tag{2}$$

Five sigma is a declared projected coverage metric, not complete containment of an unbounded Gaussian. Beta reduction in m must not be called a clearance increase in m. Historical optics-only ensembles use a separate conservative $100\,\mathrm{nm\,rad}/10\,\mathrm{nm\,rad}$, 0.11 % beam configuration and are not pooled with this study.

At 4 GeV the rigidity is approximately $13.342\,56\,\mathrm{T\,m}$. For K in m^{-2}, $G = K(B\rho)$ in $\mathrm{T\,m}^{-1}$ and $B_{\mathrm{pole}} = |G|r$ in T. Assumed bounds are $|K| \le 2.5\,\mathrm{m}^{-2}$ for q11/q33 and $|K| \le 3\,\mathrm{m}^{-2}$ for the other quadrupoles. With the assumed 19.35 mm bore, the respective maxima are $33.356\,\mathrm{T\,m}^{-1}/40.028\,\mathrm{T\,m}^{-1}$ and $0.645\,\mathrm{T}/0.775\,\mathrm{T}$, below the assumed 1.2 T screening limit. This calculation checks model consistency, not magnet engineering feasibility.

2.3 Element-wise restrictions

Every element of the BTS line carries its own aperture (Table 2); no element is added, removed or subdivided, and AT checks each aperture at the element entrance. Interior envelopes are sampled for the nominal profile on an observation-only copy (four subdivisions per quadrupole, eight per bend or septum), which is never tracked. The only restriction with a source is the injection-septum blade. In the companion NKM study [4], the injected beam leaves the septum at $x = -19.478\,\mathrm{mm}$, with the stored-beam edge at $-16\,\mathrm{mm}$ and a 2 mm blade, i.e. 1.478 mm from the blade's outer face, chosen there as three rms sizes plus 1 mm. We place the blade at $x = 1.478\,\mathrm{mm}$ in the injection septum and at the line end (marker m4), on the side facing the stored beam. All other dimensions are declared assumptions and are swept. Bend chambers are taken as curved: the sagitta of the reference arc over its chord is 30.8 mm in b2 and 19.5 mm in b1/b3, so a straight chamber would need that much extra width. Clearances (Eq. 2) are evaluated at every aperture, with the projected halfwidth of a round aperture reduced by the other plane's centroid.

Table 2: Element-wise aperture model (halfwidths; for the injection septum, a_x is the blade distance on the $+x$ side and the outer wall is at $-8.5\,\mathrm{mm}$). Only the blade position follows a source; all other values are assumptions.

Region	Shape	a_x [mm]	a_y [mm]	Basis
Drifts	round	19.350	19.35	assumed, equal to bore
Quadrupoles q11–q33	round	19.350	19.35	assumed bore
Extraction kicker	rect.	15.000	10.00	assumed
Extraction septum	rect.	7.500	5.00	assumed; swept
Bends b1–b3 (curved)	rect.	30.000	10.00	assumed; gap swept
Injection septum	one-sided	1.478	5.00	blade from companion

2.4 Errors, jitter and trajectory correction

Static machine errors are independent Gaussian quadrupole gradient errors (0.1 % rms), x/y offsets (100 µm), rolls (0.5 mrad) and dipole field errors (0.05 %). Each shot adds booster jitter at m1: entrance beta factors (5 %), centroid position 0.5 mm, angle 0.2 mrad and momentum 0.1 %, the latter carried by the entrance dispersion. These levels are modeling assumptions, with the same booster jitter as the companion NKM study [4]; their sensitivity is examined in Section 4.5.

Static trajectory errors are corrected without changing the line: no corrector or monitor element is added. The horizontal knobs are field trims of the six existing dipoles, applied as additional fixed-geometry dipole fields like the dipole errors. Horizontal readings are taken at existing element boundaries, after each quadrupole, at the injection-septum entrance and at the line end, with a static 50 µm rms reading error (offset and calibration residual). The trims minimize the weighted readings (septum entrance and end weighted 3) with the response of the error-free model line, Tikhonov-regularized and limited to ± 1 mrad of kick, in two passes. The correction uses no knowledge of the actual errors and does not act on shot jitter. The line has no vertical knob, so the vertical plane is not corrected. Every machine is evaluated uncorrected and corrected on identical particles.

3 Physical criterion, validation and comparison protocol

3.1 Active restrictions and boundary-fixed clearances

The two end stations sit where the optics are fixed by boundary data: the entrance Twiss functions and dispersion are inputs, and every admissible setting matches the exit to its target. For a centered nominal beam the clearance there is therefore $c_u = a - 5\sqrt{\epsilon_u\beta_u + D_u^2\sigma_\delta^2}$ with the boundary values of Table 1, for every quadrupole setting K that satisfies the match. With $a = 19.35\,\mathrm{mm}$ this gives $17.455\,\mathrm{mm}/18.865\,\mathrm{mm}$ at the entrance and $18.555\,\mathrm{mm}/19.064\,\mathrm{mm}$ at the exit (x/y), exactly the values found numerically for both settings in the four-station model (Supplement). The two interior stations have 30 mm halfwidths and could limit only if a five-sigma size exceeded about 12.5 mm, which no matched setting approaches. In this four-station model the nominal minimum clearance is thus analytically independent of K: the "saturation" of the minimum is a consequence of placing restrictions only where the optics are fixed, not a finding about the optics. Under errors, the only K-dependent part of the end-station clearances is the response of the centroid at the exit to errors and jitter. The same holds for any restriction upstream of q11 or downstream of q33: there the optics are fixed by the boundary data, so only the trajectory response can depend on K.

At a single symmetric active station away from $\bar{u} = 0$, the local sensitivity is

$$\frac{\partial c_u}{\partial K_j} = -\operatorname{sgn}(\bar{u})\frac{\partial \bar{u}}{\partial K_j} - \frac{5}{2\sigma_u}\frac{\partial \Sigma_{uu}}{\partial K_j}. \tag{3}$$

Switching restrictions or crossing zero centroid makes this expression piecewise. In an uncoupled plane $\Sigma_{uu} = \epsilon_u\beta_u + D_u^2\sigma_\delta^2$; our coupled calculation retains the complete covariance. Reducing beta controls one size contribution, not necessarily dispersion or orbit sensitivity. We export these variances and centroids separately, adding variances rather than RMS sizes.

3.2 Jitter transfer between matched ends

Because both ends are matched, the horizontal transfer matrix from m1 to m4 is

$$M_x = B_4 R(\mu_x) B_1^{-1}, \qquad B = \begin{pmatrix} \sqrt{\beta} & 0 \\ -\alpha/\sqrt{\beta} & 1/\sqrt{\beta} \end{pmatrix}, \tag{4}$$

with $R(\mu_x)$ a rotation by the horizontal phase advance and $B_{1,4}$ the boundary Twiss values of Table 1. A shot centroid at m1 with covariance J (position 0.5 mm, angle 0.2 mrad) therefore reaches the exit with covariance $M_x J M_x^{\mathsf{T}}$ plus the momentum term $\sigma_{\delta,j}^2 \mathbf{d}_4\mathbf{d}_4^{\mathsf{T}}$, which is the same for every matched setting. The quadrupoles influence the exit jitter only through μ_x. Over a full period the exit position spread ranges from 0.26 mm to 0.95 mm and the angle spread from 0.12 mrad to 0.41 mrad, in opposite phase (Fig. 4, left): a matched line can move booster jitter between position, which the septum blade sees, and angle, which the injection kicker sees, but cannot remove it. Steered static errors add a small residual (Section 4.2).

3.3 Loss-based observable and design

For a Gaussian projection with centroid $\bar{u}$ and rms size σ at an aperture with edges $a_\pm$, the lost fraction is

$$P = \Phi\left(\frac{a_- - \bar{u}}{\sigma}\right) + \Phi\left(\frac{\bar{u} - a_+}{\sigma}\right), \tag{5}$$

with Φ the standard normal distribution function. Per shot, the line loss is bracketed by the largest single-aperture P and the union bound; we report the former as the prediction. When the centroid is itself Gaussian, as for linearly transported jitter, the shot-averaged lost fraction is Eq. (5) with the summed covariance of beam and jitter. This makes a smooth, loss-relevant

objective. The loss-based setting minimizes the logarithm of the largest shot-averaged P on the error-free line, subject to the six exit equalities, horizontal peak beta $\leq 60\,\mathrm{m}$ (16 subdivisions, checked at 64) and the strength bounds, with SLSQP (200 iterations, tolerance 10^{-10}) from the reference, the refined low-beta setting and the best member of a bounded multistart pool. The setting is frozen before a fresh test on 200 steered machines (seeds 92001–92004), evaluated with tracked particles.

3.4 Accuracy and spatial observation resolution

Exact independent hard-edge maps are checked against AT tracking at every element boundary. The original ten-step integrator misses its 2×10^{-6} nominal maximum-entry tolerance. Twenty/forty/eighty steps give discrepancies 2.53×10^{-7}, 1.58×10^{-8} and 9.89×10^{-10}; new studies use forty. Transverse symplectic discrepancy is 1.22×10^{-15}, without a longitudinal six-dimensional certificate. Translations, rolls and fixed-geometry dipole-field perturbations are exercised independently. Momentum errors enter particle δ without double rescaling magnet fields.

The original affine error maps omit chromatic derivatives about displaced trajectories. Comparing all 600 original shots with AT Jacobians expanded about the actual centroid changes station clearances by up to 78.43 µm, but shifts the paired mean by only $-0.155\,\mu\mathrm{m}$. Eight reference and three alternative active-station classifications change. The original mean 14.61 µm and interval $[-17.91, +48.32]$ become 14.45 and $[-17.88, +48.10]$; the unresolved-benefit conclusion persists. New comparisons use centered AT maps. Moment probes are uncensored by aperture limits, so propagated moments describe the incoming distribution rather than a survivor-biased subset.

Representative simultaneous-error cases compare probe steps 10^{-6}–10^{-8} and forty/eighty integration steps. Their clearance changes are below $5 \times 10^{-6}\,\mu\mathrm{m}$. Product Gauss–Hermite quadrature at orders three, five and seven propagates the unrestricted Gaussian through nonlinear AT; five/seven convergence is negligible on the checked cases. Centred-linear versus nonlinear-moment discrepancy is at most 0.606 µm on four original extreme or representative shots and 0.6858 µm on two new machines for every regime/design. These finite-case checks are smaller than the reported effect intervals and illustrative 0.1 mm decision scale; they are not universal error bounds or Gaussian loss-tail certificates. Intentional scraping verifies first-loss accounting. Detailed checks are supplied in the supplement/evidence.

Spatial observation is refined independently of integration steps. Exact maps sample each finite element at 8, 16, 32 and 64 subdivisions, applying bend edges only once. Original reference boundary peaks 31.44 m/50.75 m become 31.63 m/51.27 m; old low-beta peaks 33.58 m/28.18 m become 33.77 m/28.50 m. Reoptimizing vertical peak beta on sixteen subdivisions converges in 34 SLSQP iterations. The resulting 64-subdivision peaks are 33.79 m/28.44 m; changes from 32 subdivisions are below 0.000 35 m. These are spatially converged sampled maxima, not exact analytical extrema or a global optimum. q32 remains at its assumed lower bound $-3\,\mathrm{m}^{-2}$.

4 Results

4.1 Active restrictions of the element-wise line

Figure 2 shows the reference and refined low-beta optics, and Fig. 3 the nominal five-sigma clearance at every aperture. The active restriction is the injection-septum channel at the blade: 0.648 mm at the septum entrance and 0.683 mm at its exit, identical for both settings because the septum lies downstream of q33. The next restrictions are the extraction septum (4.44 mm vertically, 5.73 mm horizontally), again upstream of every quadrupole. The closest interior restriction is the vertical gap of b2, with 9.12 mm and 9.32 mm for the reference and low-beta settings. Sweeping the bend half-gap (7.5 mm–12.5 mm) or the extraction-septum halfwidth (5 mm–10 mm) does not change the minimum; the blade distance sets it directly (0.17 mm,

0.648 mm, 1.17 mm and 2.17 mm for 1 mm, 1.478 mm, 2 mm and 3 mm). In nominal terms the interior optics, including the 45 % lower vertical beta peak, therefore change only clearances of order 9 mm or more; whatever the quadrupoles can do for the limiting restriction must act through the response of the trajectory to errors and jitter.

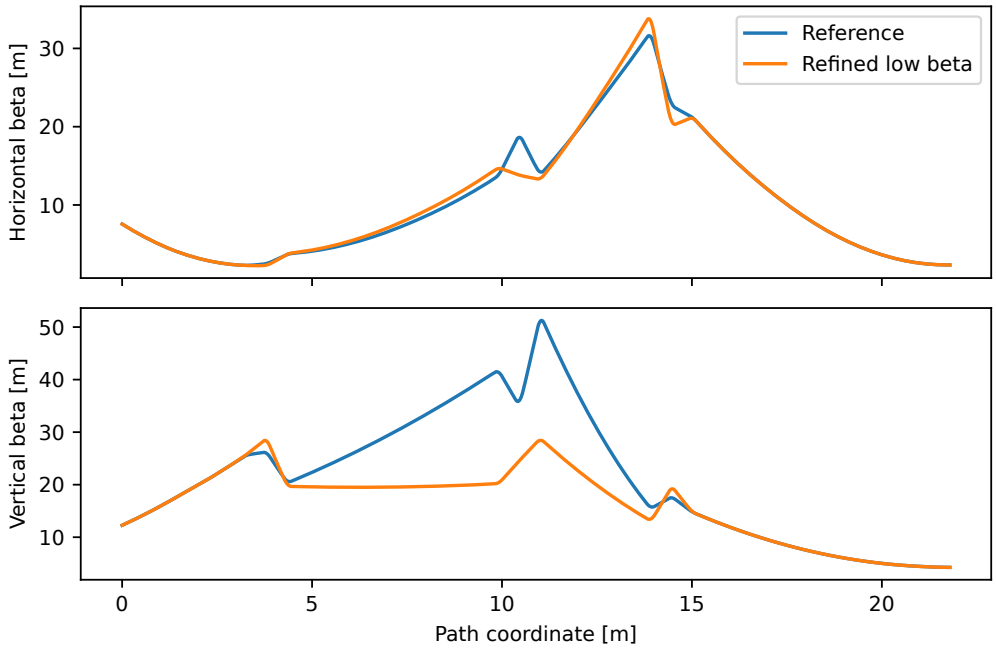


Figure 2: Reference and refined low-beta profiles drawn with Accelerator Toolbox linear-optics plotting, using the entrance Twiss parameters and 64 spatial subdivisions per finite element. The exit match remains fixed; the vertical improvement also changes horizontal transport.

4.2 Trajectory correction

Table 3 compares the uncorrected and corrected lines on 200 machines with three shots of 2 000 particles each (seeds 91001–91004); both settings and both variants see the same machines, shots and particles. Without correction the static errors displace the exit centroid by 1.2 mm rms, and 10.08 % of the particles are lost, almost all on the blade. With the dipole trims the exit position spread falls to 684 µm and the angle spread to 367 µrad, and the loss to 1.68 % (1.47 % for the low-beta setting); the rms trim is 204 µrad. Two trims close to the exit, b3 and the injection septum, steer both position and angle at the septum. The uncorrected vertical orbit stays far inside the vertical restrictions.

Table 3: Steering on 200 machines (3 shots, 2 000 particles each). $\bar{c}_{\min}$ is the mean minimum five-sigma clearance, Lost the tracked particle-loss fraction, Predicted the mean Gaussian loss probability of the most loss-prone aperture, Blocks the number of machines with any observed loss, and x_{rms}, x'_{rms} the rms exit centroid over shots.

Setting	Steering	$\bar{c}_{\min}$ [mm]	Lost [%]	Predicted [%]	Blocks	x_{rms} [mm]	x'_{rms} [mrad]
Ref.	none	0.584	10.08	10.04	74	1.20	0.36
Low beta	none	0.556	10.00	10.01	79	1.20	0.38
Ref.	dipole trims	0.524	1.68	1.68	47	0.68	0.37
Low beta	dipole trims	0.523	1.47	1.47	44	0.68	0.38

Two observations matter for the design question. First, after correction the exit centroid spread,

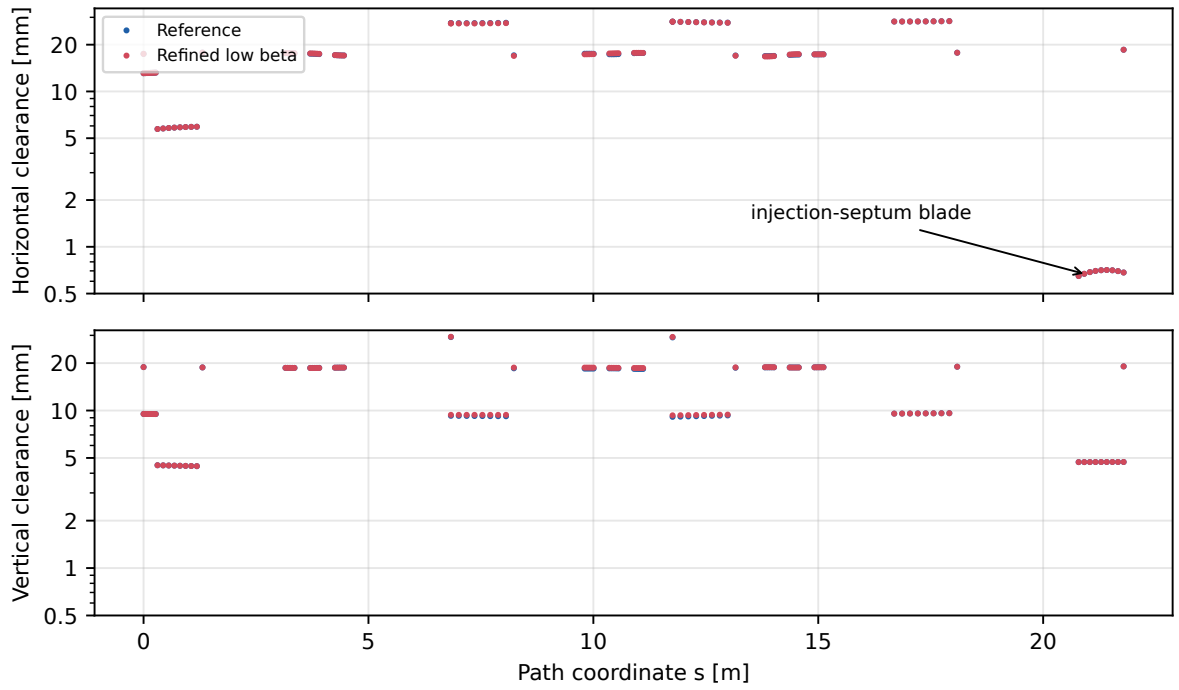


Figure 3: Nominal five-sigma clearance at every aperture of the element-wise line (logarithmic scale), horizontal (top) and vertical (bottom), for the reference and refined low-beta settings. The two settings coincide at both septa; they differ only where clearances exceed 9 mm. Design beam, no errors.

about 0.68 mm and 0.37 mrad, is mainly that of the booster jitter, and it sets the remaining blade loss. Second, the mean minimum clearance does not rank the variants by loss: the uncorrected line has the largest mean clearance (0.584 mm against 0.524 mm) and six times the loss, because a static offset away from the one-sided blade raises the mean while offsets towards it are lost. The Gaussian loss prediction, by contrast, agrees with the tracked loss within 0.4 % (relative) in every row. The reference and low-beta settings are indistinguishable in every column.

4.3 Reachable phase advance and the loss-based setting

Bounded least-squares matching from 360 random starts finds 349 matched settings within the strength bounds. Their horizontal phase advance spans only 200.2°–232.0°: the exit match, with its dispersion targets and the bend layout, pins μ_x to an interval in which booster jitter falls largely on the exit position. The reference (211.8°) and low-beta (210.8°) settings lie near its middle. Within the interval the predicted shot-averaged blade loss varies by more than a factor of ten (Fig. 4, right) and decreases monotonically towards the lower edge.

The loss-based search ends at that edge, $\mu_x = 200.2°$. All three SLSQP starts reach the 200-iteration limit; only the start from the best pool member is feasible, and it improves that member by 1.0 %. The setting is therefore a feasible point at the boundary of the reachable interval, not a certified optimum. Its predicted shot-averaged loss is 0.27 % against 1.28 % for the reference, because its exit position jitter is 506 instead of 643 µm (Table 4).

On the fresh machines the tracked loss is 2.30 %, 2.27 % and 0.69 % for the reference, low-beta and loss-based settings, each within 1 % of its prediction. The loss-based setting changes the loss by -1.61 percentage points against the reference (pointwise 95 % machine-cluster interval [-2.33, -0.96]), a factor of 3.3; low beta changes it by -0.03 [-0.23, +0.17], unresolved. Its mean minimum clearance changes by only −0.017 mm [-0.035, +0.001], also unresolved: the mean-

Table 4: Frozen settings and their fresh test (200 steered machines, 3 shots, 2 000 particles each). μ_x: horizontal phase advance; $\hat{\beta}$: larger peak beta; $\sigma_{x,j}$, $\sigma_{x',j}$: exit jitter from Eq. (4); $\bar{P}$: shot-averaged loss on the error-free line; Lost and Pred.: tracked and predicted loss on the test machines; Blocks: machines with any loss.

Setting	μ_x [deg]	$\hat{\beta}$ [m]	$\sigma_{x,j}$ [mm]	$\sigma_{x',j}$ [mrad]	$\bar{P}$ [%]	Lost [%]	Pred. [%]	Blocks
Ref.	211.8	51.3	0.643	0.327	1.28	2.30	2.31	57
Low beta	210.8	33.8	0.631	0.331	1.16	2.27	2.28	55
Loss-based	200.2	48.1	0.506	0.368	0.27	0.69	0.69	29

clearance metric cannot resolve this loss reduction. The price is an exit angle jitter larger by 13 % (368 against 327 µrad), which the injection kicker must accept. The loss is concentrated in shots whose entrance jitter is two to three standard deviations: the five worst of the 200 reference test machines contribute 38 % of its loss, the worst losing 43 % per shot on average. The machine-cluster intervals account for this heavy tail, and the population mean differs between independently seeded populations (Section 4.5).

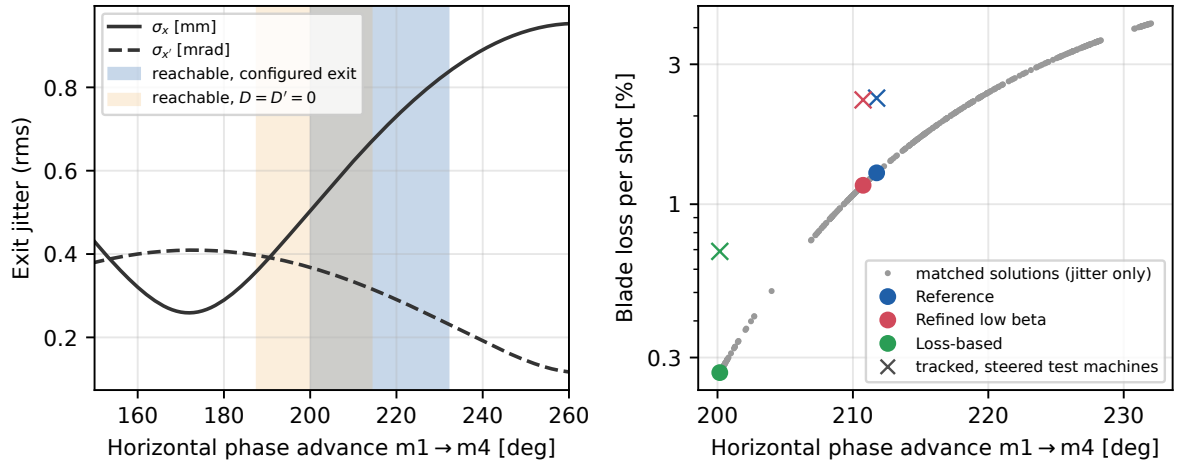


Figure 4: Left: rms exit position (solid, mm) and angle (dashed, mrad) jitter from Eq. (4) versus horizontal phase advance, with the reachable intervals for the configured exit (blue) and a dispersion-free exit (orange; its momentum term differs slightly). Right: predicted shot-averaged blade loss of every matched solution (grey) and of the three settings (circles), with the tracked loss on steered test machines (crosses), which also include static residuals.

4.4 Exit target and dispersion

At m4 the configured target gives the injected beam an rms divergence of 76 µrad: 57 from the betatron emittance and 50 from $D'_x\sigma_\delta$. The 0.1 % shot energy jitter adds a centroid angle of 47 µrad and a position of 81 µm rms. Both terms are fixed by the target, the same for every matched setting, and absent for a dispersion-free exit.

Matching the same Twiss functions with $D_x = D'_x = 0$ at m4 is feasible within the strength bounds: all 115 solutions from 120 bounded random starts keep both peak betas below 60 m (smallest larger peak 26.3 m). The dispersion-free target also widens the reachable phase advance to 187.7°–214.2° (Fig. 4), which includes phases where booster jitter falls mainly on the exit angle. The best of these solutions ($\mu_x = 187.7°$, peak betas 31.4 m/48.2 m) has a shot-averaged blade loss of 4.1×10^{-5}, 65 times below the loss-based setting with the configured target. Its exit jitter is 350 µm and 394 µrad, and the beam divergence falls to 57 µrad. These numbers are

analytic, from the same Gaussian model whose predictions agreed with tracking within 1 % above; the dispersion-free settings were not tracked on test machines.

The exit target is thus the BTS design quantity with the largest leverage on injection. Removing the exit dispersion lowers the injected beam's divergence by a quarter and lets the line put booster jitter where the septum blade does not see it, but the angle jitter, 394 µrad against 327 for the reference, then arrives at the injection kicker. In the companion study [4], the kicker's 80 % angular acceptance is 0.44 mrad, and once the static vertical injection error left by transfer-line magnet errors is removed, booster position and angle jitter are the dominant kicker-specific capture losses. This line has no vertical correction; correcting that vertical error will be addressed in a future study. Whether the trade favors the blade or the kicker must be decided with the ring, not within the line.

4.5 Sensitivity to the assumed error levels

The 0.5 mrad rms quadrupole roll is large for a transfer line, and 0.1 % rms shot energy jitter is at the high end of booster performance. Table 5 repeats the steered evaluation with rolls of 0.1 mrad, 0.2 mrad and 0.5 mrad and energy jitter of 0.05 % and 0.1 % on another 200 independently seeded machines (analytic loss and clearance, no particle tracking). The roll level changes the predicted loss by at most 0.000 percentage points: rolls couple a small part of the horizontal motion into the vertical plane, whose restrictions keep several millimeters of clearance, and leave the horizontal blade loss unchanged. Halving the energy jitter lowers the reference loss from 1.34 % to 1.32 %, because its 81 µm exit contribution is small compared with the betatron jitter. The ranking of the settings is the same at every level. On this population the loss-based setting is 5.6 times below the reference, against 3.3 on the tracked test population: the size of the gain depends on the sampled machines, as expected for a heavy-tailed loss, while its sign does not.

Table 5: Sensitivity of the steered line to the roll and energy-jitter levels: mean predicted loss and mean minimum five-sigma clearance on 200 machines with three shots each (seeds 93001–93004; analytic, not tracked).

Roll	$\sigma_{\delta,j}$	Predicted loss [%]			$\bar{c}_{\min}$ [mm]	
[mrad]	[%]	Ref.	Low beta	Loss-based	Ref.	Loss-based
0.1	0.05	1.32	1.20	0.23	0.519	0.504
0.1	0.10	1.34	1.24	0.24	0.517	0.502
0.2	0.05	1.32	1.20	0.23	0.519	0.504
0.2	0.10	1.34	1.24	0.24	0.517	0.502
0.5	0.05	1.32	1.20	0.23	0.519	0.504
0.5	0.10	1.34	1.24	0.24	0.517	0.502

5 Discussion and conclusions

In a BTS line matched at both ends, the beam size at the ends is fixed and so is the nominal clearance of every restriction outside the quadrupole section. With the companion blade geometry the injection septum is that restriction, and it limits the line by an order of magnitude more than any interior aperture. Lowering the interior peak beta, the usual design heuristic, therefore changes only clearances of several millimeters and, on fresh steered machines, leaves the loss unchanged.

The quadrupoles act on the loss through the trajectory. Static errors are removed by field trims of the existing dipoles, without adding any element to the line; the two trims nearest the exit steer both position and angle at the septum. What remains is booster shot jitter, and for matched ends its arrival at the exit depends only on the horizontal phase advance. The line

chooses whether the jitter appears as position at the blade or as angle at the injection kicker, but it cannot remove it. For the configured exit, matched solutions reach only a 32° interval of phase advance, and the best setting in it reduces the blade loss 3.3-fold at 13 % more angle jitter. A dispersion-free exit widens the interval and lowers the predicted loss much further, again at the cost of angle jitter. The choice between blade loss and injection-angle jitter belongs to the injection design as a whole: for the companion kicker, whose 80 % angular acceptance is 0.44 mrad, exit-angle jitter of 0.33 mrad–0.39 mrad is of the same order, so the trade must be evaluated with ring capture, and the booster's angle stability bounds what either choice can achieve.

Two methodological points carry over. The mean minimum five-sigma clearance does not rank settings by loss: it rewards static offsets away from a one-sided restriction and is blind to the jitter tail. A Gaussian loss probability from the propagated centroid and covariance agrees with tracked losses within 1 %, is smooth, and can be optimized directly. The phase-advance view also gives a quick test for any matched transfer line: compute μ_x, read off where the booster jitter lands, and compare with the restriction and the kicker acceptance.

All aperture dimensions except the blade position, the choice of dipole trims and monitor points for steering, and the error levels are declared assumptions; sweeps and a sensitivity study bound their influence on these conclusions. The loss is heavy-tailed over machines and its population mean varies between independently seeded populations, while the ranking of settings does not. The dispersion-free results are analytic. This paper neither certifies real transmission nor derives construction tolerances, ring capture or kicker transparency; the four-station analysis and historical statistics are preserved in the supplement.

Statements and Declarations

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Generative assistance. Generative AI assistance (OpenAI Codex and Anthropic Claude through Claude Code) was used for software, numerical execution, literature checking and manuscript drafting beyond copy editing, as in the companion NKM study. The author must validate and approve the final work.

Data and code availability. The simulation code, configurations and analysis scripts are available at `https://github.com/cspark7701/nkm-injection`. The selected raw simulation data, exact generating-source snapshots, environment identities, SHA-256 hashes and reproduction commands are provided to the referees as a supplementary attachment and are available from the author on reasonable request. No public data deposit has been made.

A Evidence and supplementary material

The element-wise aperture study (element-wise apertures, steering, phase-advance pools, loss-based setting, fresh test, exit target and sensitivity) is in `results/bts_journal_campaign_r3b`

_20261015, one directory per stage, each with its configuration, source hashes and the protocol (round3_protocol.json, which also records the superseded corrector layouts). The supplement preserves the four-station study, historical matching/optical statistics, block-loss inference and numerical limitations. Reviewed and original campaign identities, all previous reports, new frozen designs, station decompositions and point-by-point responses are included in the local attachment. The separate nominal phase-space diagnostic supplies initial and final particle coordinates, geometric slopes, seed and generating-source identity. Each numerical stage records source hashes and uses a fresh result directory. The first invalid shared-aperture run is retained as a failure record, not selected evidence. Technical validation does not imply journal acceptance.

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